1 Literature

Fama and MacBeth, 1973, Ma et al., 2020. Fama French 3 and 5 factors. GMM 1982. Ghysel's critique of the dynamic CAPM. Also should look at the local linear and local maximum likelihood estimation methods. Jagannathan and Z. Wang, 1998b provides a good reading on the subject. Shanken, 1992a on the standard error of the Fama-Macbeth Regression.

For example, the CAPM assumes one factor that is the market excess return r_m.

$$E(r_i) = \beta_i E(r_m) \quad \text{where} \quad \beta_i := \frac{\text{cov}(r_i, r_m)}{\text{var}(r_m)}$$

Fama and MacBeth, 1973 derive testable restrictions from the CAPM model and embed the model in a linear regression:

$$r_{it} = \gamma_{0t} + \gamma_{1t}\beta_i + \gamma_{2t}\beta_i^2 + \gamma_{3t}s_i + \epsilon_t$$

where β_i is estimated with $\hat{\beta}_i$ from time series regression for each i

$$r_{it} = \alpha + \beta_i r_{mt} + u_{it}$$

The testable hypotheses are

1. $E(\gamma_0) = 0$ there is no pricing error in the expected excess return.
2. $E(\gamma_2) = 0$ the linearity test.
3. $E(\gamma_3) = 0$ there is no other risk measure.
4. The Sharpe-Lintner Hypothesis that $E(\gamma_1) = R_{ft} > 0$.

Ma et al., 2020 considers Gibbons, Ross, and Shanken, 1989 tests with large N and points out that the factor loadings change substantially over time, they conduct a High-dimenisonal alpha test and

Jegadeesh et al., 2019 and Chaieb, Langlois, and Scaillet, 2018 are new papers that referenced the Jagannathan and Z. Wang, 1998b, see also Jagannathan and Z. Wang, 1998a.

Generally, people test CAPM or other asset pricing models by testing that the $\alpha = 0$ (Jensen's α), meaning there is no systematic pricing error(in conditional expectation).

K. Q. Wang, 2003 does a testing of CAPM and factor models in a nonparametric setting with conditioning information. The starting point is the SDF characterisation of

excess returns r_i:

$$E_t(m_{t+1}r_{i,t+1}) = 0 \implies E_t\left(\left[1 - r_{m,t+1}\frac{E_t(r_i, r_m)}{E(r_m^2)}\right]r_{i,t+1}\right) = 0 \tag{1}$$

Then $m_{t+1} = 1 - b(x)r_{m,t+1}$ is nonparametrically estimated from the market excess return $r_{m,t}, t = 1, \ldots, T$. The covariates x is four-dim in the empirical part. Then they test relationship by the following regression(they use weighted Least Square for technical reason).

$$\hat{m}_{t+1}r_{m,t+1} = z_t'\delta_i + u_{i,t+1} \quad \text{where} \quad z_t \in I_t$$

For testing a economic theory in a nonparametric setting, see Matzkin, 1994.

For *error-in-variable* problem, there is an Fan and Truong, 1993 on nonparametric regression with error in variables, they proposed deconvolution method to account for the observation errors. There is Bickel and Ritov, 1987 on efficient parametric regression: there is identification problem in the error-in-variable problem, which serves as an important example in Rothenberg, 1971. Reiersøl, 1950 is a paper on the identification issue, it shows equivalence with the assumption of normal measurement error and show equivalence using characteristic functions, this method should be noted.

For *cross-sectional correlation problem*, we can see Li and Racine, 2007 chapter 18 and also Robinson, 2012.

Acharya and Pedersen, 2005 proposes a liquidity factor and uses the cross-sectional method to verify. Further applications, see Jegadeesh et al., 2019 page 2. Jegadeesh et al., 2019 proposes an IV method to deal with the first step estimation error, or error in variable, this allows them to use individual stocks instead of portfolios, they do the test on CAPM **and** factor models.

Apparently Fama Macbeth didn't give statistical reasons for the two-pass regression procedure. Gibbons, Ross, and Shanken, 1989, Gibbons, 1982 and Shanken, 1992a should be necessary companion.

Shanken, 1992a mentioned that the analysis of MLE is not easy because they have N, T both $\to \infty$. The underlying model is richer than just simply APT, but see Shanken 1982 1985 1992 on APT and Dybvig and Ross, 1985 for discussion.

Shanken, 1992a gives a general parametric framework where he distinguished the factors that are and are not returns, and derive asymptotic distribution of the two-pass estimator of the γ's. The first lemma regards the first step estimation.

See Ross for APT. Dybvig and Ross, 1985 says that APT theory incoporates assumptions other than a simple factor generating structure so any test of APT should

take account of that. Chamberlain, 1983 is also a must-read.

Herskovic, 2018 referneced the Shanken, 1992a.

There is a different paper Shanken, 1992b. For the Eubank and Spiegelman, 1990 might be useful for the inference. Hardle and Mammen, 1993.

Follow the line of specification test as in Hardle and Mammen, 1993, there is a paper Chen, Gao, and Tang, 2008 which gives an extensive review of relevant literature. Two of the papers are worth mentioning, the genralized likelihood method and the test of $\hat{\epsilon} = 0$. But it requires some ordering of the observations, which seems a bit odd. The empirical likelihood method and generalized likelihood ratio test seems to be important in nonparametric inference literature, should check it out. Fan and Zhang, 2004 puts an even more general likelihood ratio where the error term is no longer required to be iin a parametric family.

There seems to be a lot of specificaton tests. and Fama-Macbeth is quite popular, Feng, Giglio, and Xiu, 2017 is an example, where they use double Lasso selection to both select the relevant factor and keep some of irrelevant factors in the regression to avoid the omitted variable problem and then they use Fama-macbeth to evaluate the validity of the factors.

Hong and White, 1995 is a specification test with series regression.

Fan and Yao, 2008 also has a chapter on generalized likelihood ratio statistic.

Apparently **horowitzAdaptiveRateOptimalTest2001** is an important article, it's also mentioned in Li and Racine, 2007.

Here the econometric problem is to extend the GLR test to incorporate first-step estimation error.

2 Model Set-up

Let $R_t := (R_{1,t}, \ldots, R_{Nt})_{t=1}^T$ be observed returns on assets $1, \ldots, N$. With SDF representation:

$$E(R_{t+1} m_{t+1}) = 1$$

We want to test the hypothesis that k-dimenisonal observed $F_t := (F_{1,t}, \ldots, F_{k,t})_{t=1}^T$ are pricing factors in the sense that:

$$m_t = \lambda_0 + F_t' \lambda \tag{2}$$

which will translate into a beta pricing model(Chapter 6 of Cochrane, 2001):

$$E(r_i) = \beta_i' \gamma_1 \quad \forall i \tag{3}$$

where $r_{it} = R_{it} - R_f$ is the excess return, β_i is the following regression coefficient of asset i on the factors F_t in

$$r_{it} = a_i + F_t' \beta_i + \epsilon_{it} \tag{4}$$

The relationship we want to test is given by Equation 3. We embed this relationship in the following model:

$$E(r_i) = g(\beta_i) + s_i' \gamma_2 \tag{5}$$

where s_i is a vecotr of other candidate measure of risk.

The testable restrictions we impose are:

1. $g(0) = 0$;
2. $g(x) = x' \gamma_1$;
3. $\gamma_2 = 0$.

Let $\theta = [a', B, \gamma_2', g(\cdot)]$ be our parameters, where $B' = \begin{bmatrix} \beta_1 & \beta_2 & \dots & \beta_n \end{bmatrix}$ we can estimate them in a GMM framework:

$$\Psi(\theta) = \Big[E(r_t - a - BF_t) \Big] \tag{6}$$

Under usual conditions, we have the asymptotic normality of the $\hat{\beta}_i$ that comes from the time series regression so that

$$\hat{\beta}_i = \beta_i + O_p(\frac{1}{\sqrt{T}})$$

In the second step we have a regressor that's contaminated by estimation error, and we want to estimate the function $g(\cdot)$ with the intercept included in the $g(\cdot)$ function, the regression model is for $i = 1, \dots, N$

$$r_{it} = g_t(\beta_i) + \epsilon_{it}$$

we want to establish asymptotic normality of the kernel estimator, there might be dynamic heteroscedasticity in $\epsilon_{i,t}$

3 Estimation

First we run the time-series regression:

$$R_i = F\beta_i + \epsilon_i \quad \text{for} \quad i = 1, \ldots, N$$

where F is $T \times k$ and $R_i : T \times 1$. Denote $\beta = (\beta_1', \ldots, \beta_N')'$ be $N \times k$.

Then $\hat{\beta}_i = (F'F)^{-1}F'R_i = \beta_i + (F'F)^{-1}F'\epsilon_i$.

In the cross-sectional Regression, we have regression:

$$(\hat{g}_t(x), \sim) = \arg\min_{a,b} \sum_i \left(R_{it} - a - \hat{\beta}_i' b\right)^2 K_h(x - \hat{\beta}_i) \quad \text{for} \quad t = 1, 2, \ldots, T$$

Then we can construct the **Generalized Likelihood Ratio Statistic** as follows:

Proposition 1. *Under regularity conditions, we have*

$$\sqrt{T}\left(\hat{\beta}_i - \beta_i\right) \rightsquigarrow N(0, \Sigma_\epsilon \otimes \Sigma_F^{-1})$$

3.1 Semiparametric Cross-sectional Regression

Our regression model is

$$r_i = g(\beta_i) + \gamma_i' s_i + \epsilon_i$$

where s_i are a vector of other risk measures.

I think some work need be done for the estimation of the covariance because there seem to be cross-sectional correlation. We can show asymptotic normality, see Li and Racine, 2007 chapter 2 and 18, should also check chapter 19 and 20. Then we follow Hardle and Mammen, 1993 to see whether the linearity holds, which proposes a L^2 test and suggest bootstrap method. For that purpose, we need to find uniform convergence rate, see Mack and Silverman, 1982, which is based on Komlós, Major, and Tusnády, 1975, see Berkes, Liu, and Wu, 2014 for the extension to dependent data. Komlós, Major, and Tusnády, 1975 states a result due to Bartfai 1966, and later a simple proof for that is proved in ERD, 1970.

As pointed out by Fama and MacBeth, 1973, we should calculate that $Eg(\cdot)$ is a linear function rather then $g(\cdot)$ being linear.

Okui and Yanagi, 2019 provides a kernel density estimation method with the first-step estimation error. Dhaene and Jochmans, 2015 provides a way to correct for biases.

Fama and Macbeth mentioned that the estimated $\hat{\beta}_i$ tends to be above the true β_i for large β, I wonder why that's the case.

Later we can try this technique on a subset of the assets and see whether it works, sort of like a bootstrap.

Cochrane, 2001 provides a way to put the two passes in one pass with GMM framework. Tao, 2020 might also be helpful.

$$R_{it} = h_t(\beta_i) + e_{it}$$

where we estimate β_i from the time series of returns. Then each period we have h_t is only driven by the cross-section size and the time series estimation error in β_i. We want to test that $h_t(b) = b * R_m t$ for some observed $R_m t$ or even some latent market factor so $b * U_t$, since then we would have $h_t(b)/h_s(b) = U_t/U_s$ which does not vary with b, which would be testable

4 Questions

1. Test for nonlinearity in the beta: should we use sieve; should we use daily data of each stocks; the error-in-variable problem. Can we put it in the SDF framework.

2. inner-product structure fits the real data, and so the arbitrage pricing theory implies linear pricing function in that space?

3. The eigenvectors seem to be stable, can it be a random thing?

4. Can we extract the signal from the time-varying betas.

5. About the exogeneity of the market portfolio, does it make sense to exclude large-value stocks so that the weights are small.

6. General question: is there a similarity between time series and the high-frequency literature.

5 Data

We collect data from CRSP, 1960-2019, all stocks with NaN are discarded, leaving with 304 stocks. Use window length of 100 days, 50 days overlapping. Calculate the beta in the first window and sort the stocks into 31 portfolios. Then calculate the betas in each window and the return in that period,

References

Acharya, Viral V. and Lasse Heje Pedersen (Aug. 1, 2005). "Asset Pricing with Liquidity Risk". In: *Journal of Financial Economics* 77.2, pp. 375–410. ISSN: 0304-405X. DOI: 10.1016/j.jfineco.2004.06.007. URL: http://www.sciencedirect.com/science/article/pii/S0304405X05000334 (visited on 02/27/2020).

Berkes, István, Weidong Liu, and Wei Biao Wu (2014). "Komlós–Major–Tusnády Approximation under Dependence". In: *The annals of probability* 42.2, pp. 794–817.

Bickel, P. J. and Y. Ritov (1987). "Efficient Estimation in the Errors in Variables Model". In: *The Annals of Statistics*, pp. 513–540.

Chaieb, Ines, Hugues Langlois, and O. Scaillet (June 2018). *Time-Varying Risk Premia in Large International Equity Markets*. SSRN Scholarly Paper ID 3103752. Rochester, NY: Social Science Research Network.

Chamberlain, Gary (1983). "Funds, Factors, and Diversification in Arbitrage Pricing Models". In: *Econometrica: Journal of the Econometric Society*, pp. 1305–1323.

Chen, Song Xi, Jiti Gao, and Cheng Yong Tang (Feb. 2008). "A Test for Model Specification of Diffusion Processes". In: *The Annals of Statistics* 36.1, pp. 167–198. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/009053607000000659. URL: https://projecteuclid.org/euclid.aos/1201877298 (visited on 04/14/2020).

Cochrane, John Howland (2001). *Asset Pricing*. Princeton University Press. 556 pp. ISBN: 978-0-691-07498-6. Google Books: WpmtKSoM4KcC.

Dhaene, Geert and Koen Jochmans (2015). "Split-Panel Jackknife Estimation of Fixed-Effect Models". In: *The Review of Economic Studies* 82.3, pp. 991–1030.

Dybvig, Philip H. and Stephen A. Ross (1985). "Yes, the APT Is Testable". In: *The Journal of Finance* 40.4, pp. 1173–1188.

ERD, PAUL (1970). "On a New Law of Large Numbers". In: *J. Anal. Math* 23.103111, p. 8.

Eubank, R. L. and C. H. Spiegelman (June 1, 1990). "Testing the Goodness of Fit of a Linear Model via Nonparametric Regression Techniques". In: *Journal of the American Statistical Association* 85.410, pp. 387–392. ISSN: 0162-1459. DOI: 10.1080/01621459.1990.10476211. URL: https://amstat.tandfonline.com/doi/abs/10.1080/01621459.1990.10476211 (visited on 03/01/2020).

Fama, Eugene F. and James D. MacBeth (1973). "Risk, Return, and Equilibrium: Empirical Tests". In: *Journal of Political Economy* 81.3, pp. 607–636. ISSN: 0022-3808. URL: https://www.jstor.org/stable/1831028 (visited on 10/10/2019).

Fan, Jianqing and Young K. Truong (Dec. 1993). "Nonparametric Regression with Errors in Variables". In: *The Annals of Statistics* 21.4, pp. 1900–1925. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/aos/1176349402.

Fan, Jianqing and Qiwei Yao (Sept. 2008). *Nonlinear Time Series: Nonparametric and Parametric Methods*. Springer Science & Business Media. ISBN: 978-0-387-69395-8.

Fan, Jianqing and Jian Zhang (Oct. 2004). "Sieve Empirical Likelihood Ratio Tests for Nonparametric Functions". In: *The Annals of Statistics* 32.5, pp. 1858–1907. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/009053604000000210. URL: https://projecteuclid.org/euclid.aos/1098883775 (visited on 04/15/2020).

Feng, Guanhao, Stefano Giglio, and Dacheng Xiu (2017). "Taming the Factor Zoo". In:

Gibbons, Michael R. (1982). "Multivariate Tests of Financial Models: A New Approach". In: *Journal of financial economics* 10.1, pp. 3–27.

Gibbons, Michael R., Stephen A. Ross, and Jay Shanken (1989). "A Test of the Efficiency of a given Portfolio". In: *Econometrica: Journal of the Econometric Society*, pp. 1121–1152.

Hardle, W. and E. Mammen (Dec. 1993). "Comparing Nonparametric Versus Parametric Regression Fits". In: *The Annals of Statistics* 21.4, pp. 1926–1947. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/aos/1176349403. URL: https://projecteuclid.org/euclid.aos/1176349403 (visited on 02/22/2020).

Herskovic, Bernard (Apr. 2018). "Networks in Production: Asset Pricing Implications". In: *The Journal of Finance*. ISSN: 00221082. DOI: 10.1111/jofi.12684.

Hong, Yongmiao and Halbert White (1995). "Consistent Specification Testing Via Nonparametric Series Regression". In: *Econometrica* 63.5, pp. 1133–1159. ISSN: 0012-9682. DOI: 10.2307/2171724. URL: https://www.jstor.org/stable/2171724 (visited on 04/23/2020).

Jagannathan, Ravi and Zhenyu Wang (1998a). "A Note on the Asymptotic Covariance in Fama-MacBeth Regression". In: *The Journal of Finance* 53.2, pp. 799–801. ISSN: 0022-1082. URL: https://www.jstor.org/stable/117372 (visited on 07/04/2019).

— (Aug. 1998b). "An Asymptotic Theory for Estimating Beta-Pricing Models Using Cross-Sectional Regression". In: *The Journal of Finance* 53.4, pp. 1285–1309. ISSN: 00221082. DOI: 10.1111/0022-1082.00053. URL: http://doi.wiley.com/10.1111/0022-1082.00053 (visited on 02/17/2020).

Jegadeesh, Narasimhan et al. (Aug. 1, 2019). "Empirical Tests of Asset Pricing Models with Individual Assets: Resolving the Errors-in-Variables Bias in Risk Premium Estimation". In: *Journal of Financial Economics* 133.2, pp. 273–298. ISSN: 0304-405X.

DOI: 10.1016/j.jfineco.2019.02.010. URL: http://www.sciencedirect.com/science/article/pii/S0304405X19300431 (visited on 02/22/2020).

Komlós, János, Péter Major, and Gábor Tusnády (1975). "An Approximation of Partial Sums of Independent RV'-s, and the Sample DF. I". In: *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* 32.1-2, pp. 111–131.

Li, Qi and Jeffrey Scott Racine (2007). *Nonparametric Econometrics: Theory and Practice*. Princeton University Press. 770 pp. ISBN: 978-0-691-12161-1. Google Books: BI_PiWazY0YC.

Ma, Shujie et al. (Jan. 2, 2020). "Testing Alphas in Conditional Time-Varying Factor Models With High-Dimensional Assets". In: *Journal of Business & Economic Statistics* 38.1, pp. 214–227. ISSN: 0735-0015, 1537-2707. DOI: 10.1080/07350015.2018.1482758. URL: https://www.tandfonline.com/doi/full/10.1080/07350015.2018.1482758 (visited on 02/10/2020).

Mack, Y. P. and B. W. Silverman (1982). "Weak and Strong Uniform Consistency of Kernel Regression Estimates". In: *Zeitschrift F\dbendr Wahrscheinlichkeitstheorie Und Verwandte Gebiete* 61.3, pp. 405–415. ISSN: 0044-3719, 1432-2064. DOI: 10.1007/BF00539840.

Matzkin, Rosa L. (Jan. 1994). "Chapter 42 Restrictions of Economic Theory in Nonparametric Methods". In: *Handbook of Econometrics*. Vol. 4. Elsevier, pp. 2523–2558. DOI: 10.1016/S1573-4412(05)80011-X.

Okui, Ryo and Takahide Yanagi (2019). "Kernel Estimation for Panel Data with Heterogeneous Dynamics". In: *Available at SSRN 3128885*.

Reiersøl, Olav (1950). "Identifiability of a Linear Relation between Variables Which Are Subject to Error". In: *Econometrica: Journal of the Econometric Society*, pp. 375–389.

Robinson, Peter M. (July 1, 2012). "Nonparametric Trending Regression with Cross-Sectional Dependence". In: *Journal of Econometrics*. Recent Advances in Panel Data, Nonlinear and Nonparametric Models: A Festschrift in Honor of Peter C.B. Phillips 169.1, pp. 4–14. ISSN: 0304-4076. DOI: 10.1016/j.jeconom.2012.01.005. URL: http://www.sciencedirect.com/science/article/pii/S0304407612000061 (visited on 02/24/2020).

Rothenberg, Thomas J. (1971). "Identification in Parametric Models". In: *Econometrica* 39.3, pp. 577–591. ISSN: 0012-9682. DOI: 10.2307/1913267.

Shanken, Jay (Jan. 1, 1992a). "On the Estimation of Beta-Pricing Models". In: *The Review of Financial Studies* 5.1, pp. 1–33. ISSN: 0893-9454. DOI: 10.1093/rfs/5.1.1.

URL: https://academic.oup.com/rfs/article/5/1/1/1599649 (visited on 02/22/2020).

— (1992b). "The Current State of the Arbitrage Pricing Theory". In: *The Journal of Finance* 47.4, pp. 1569–1574.

Tao, Jing (2020). "Uniform Trinity Tests for Semi/Nonparametric GMM Models". In: *Working Paper*, p. 39.

Wang, Kevin Q. (2003). "Asset Pricing with Conditioning Information: A New Test". In: *The Journal of Finance* 58.1, pp. 161–196. ISSN: 1540-6261. DOI: 10.1111/1540-6261.00521. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/1540-6261.00521 (visited on 03/20/2019).

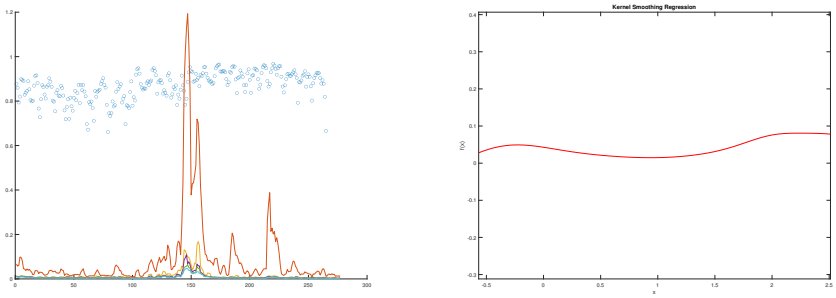

Figure 1: Kernel smoothing